QUICK SUMMARY

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JavaScript Basics:

1. Variables- var,let,const (differs by scope)
2. Data type-Number,String,Boolean,Null,Undefined,Object(complex datatype)
3. Operators-arithmetic,comparison,logical
4. Control flow- Conditional statements,loops
5. Function- set of instruction to perform a task
6. DOM manipulation- by document.getElementById();

JavaScripts Runs:

- Client side- Browser itself, provide dynamic interaction
- Server side - Node.js

Client Server Model:

- Distributed Computing model - one server can handle multiple client at a time
- Request-Response through HTTP/HTTPS(Communication Purpose)

Working:

1. User type url on web browser
2. DNS(Server) find the ip address of the website url
3. IP address sent to Client(Web browser)
4. Using ip client get html,css, js and images of the website
5. Browser render the website for the user

Request Method: [Method- URL]

- GET-retrieve data from the server
- POST-add data to the server
- PUT- modify the data in the server
- PATCH- same as PUT, but here even though update done by accessing only one attribute of the DB, other attributes won't be set as null(which is done in PUT).
- DELETE - remove a data from the DB
- HEAD- it will return only header of the request instead of returning header,http body(done in GET)
- CONNECT- establish communication between client and server through TLS/SSL (Security Purpose)
- TRACE- when client feels its request not considered by server then it echo its request and verify does its request processed and got a response or not

Response Method:

Server respond with Status code,header, body

- Status 100- Request received, on process
- Status 200-Success (general there are 200,201,202…)
- Status 300-Redirection (general)
- Status 400-Client side error
- Status 500-Server side error

Real world OOPs Models:

1. Class model- Class Diagram
2. State model- State Diagram
3. Interaction model- Use case diagram, Sequence diagram(comm flow), Activity diagram

Software Development Life Cycle:

- Design,develop and test a software
- Goal is to get a high quality software
- Satisfy user requirements

Steps:

1. Planning-define ps,project scope, identify risks with strategies
2. Defining- get all requirements-user,functional,technical (SRS)
3. Designing-with the help of SRS, design the software(DDS)
4. Developing- with the help of DDS software developed and programming tools like compiler, debugger,interpreter used here
5. Testing- software tested with SRS(whether it satisfies all)
6. Deployment- software deployed to end user and monitored

Agile:

- Agile is a Software Development Life Cycle model
- It guides the entire cycle of the software
- Instead of completing the entire steps fully to get a software (Waterfall model), it will focus on delivering smaller piece of work
- Adapt to quick changes

Scrum:

- Scrum is an agile methodology
- Good for small team works
- It has a scrum master who monitor the tasks
- Sprint planning decides what the team will work on in the following sprint
- Sprint review- review for improvisation
- Sprint retrospective- status of the project checked
- Product backlog- prioritize features

Kanban:

- Focus on real time communication and visualize the work flow
- It manages work in process, completed work and identify bottleneck

Extreme Programming:

- Release update frequently
- Done when customer feedback and fast changes are crucial